

Human antimicrobial cytotoxic T lymphocytes, defined by NK receptors and antimicrobial proteins, kill intracellular bacteria.

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Human CD8⁺ cytotoxic T lymphocytes (CTLs) contribute to antimicrobial defense against intracellular pathogens through secretion of cytotoxic granule proteins granzyme B, perforin, and granulysin. However, CTLs are heterogeneous in the expression of these proteins, and the subset(s) responsible for antimicrobial activity is unclear. Studying human leprosy, we found that the subset of CTLs coexpressing all three cytotoxic molecules is increased in the resistant form of the disease, can be expanded by interleukin-15 (IL-15), and is differentiated from naïve CD8⁺ T cells by Langerhans cells. RNA sequencing analysis identified that these CTLs express a gene signature that includes an array of surface receptors typically expressed by natural killer (NK) cells. We determined that CD8⁺ CTLs expressing granzyme B, perforin, and granulysin, as well as the activating NK receptor NKG2C, represent a population of "antimicrobial CTLs" (amCTLs) capable of T cell receptor (TCR)-dependent and TCR-independent release of cytotoxic granule proteins that mediate antimicrobial activity.

Altered effector function of peripheral cytotoxic cells in COPD.

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There is mounting evidence that perforin and granzymes are important mediators in the lung destruction seen in COPD. We investigated the characteristics of the three main perforin and granzyme containing peripheral cells, namely CD8⁺ T lymphocytes, natural killer (NK; CD56⁺CD3⁻) cells and NKT-like (CD56⁺CD3⁺) cells. Peripheral blood mononuclear cells (PBMCs) were isolated and cell numbers and intracellular granzyme B and perforin were analysed by flow cytometry. Immunomagnetically selected CD8⁺ T lymphocytes, NK (CD56⁺CD3⁻) and NKT-like (CD56⁺CD3⁺) cells were used in an LDH release assay to determine cytotoxicity and cytotoxic mechanisms were investigated by blocking perforin and granzyme B with relevant antibodies. The proportion of peripheral blood NKT-like (CD56⁺CD3⁺) cells in smokers with COPD (COPD subjects) was significantly lower (0.6%) than in healthy smokers (smokers) (2.8%, $p < 0.001$) and non-smoking healthy participants (HNS) (3.3%, $p < 0.001$). NK (CD56⁺CD3⁻) cells from COPD subjects were significantly less cytotoxic than in smokers (16.8% vs 51.9% specific lysis, $p < 0.001$) as were NKT-like (CD56⁺CD3⁺) cells (16.7% vs 52.4% specific lysis, $p < 0.001$). Both cell types had lower proportions expressing both perforin and granzyme B. Blocking the action of perforin and granzyme B reduced the cytotoxic activity of NK (CD56⁺CD3⁻) and NKT-like (CD56⁺CD3⁺) cells from smokers and HNS. In this study, we show that the relative numbers of peripheral blood NK (CD56⁺CD3⁻) and NKT-like (CD56⁺CD3⁺) cells in COPD subjects are reduced and that their cytotoxic effector function is defective.

Systematic characterization of human CD8⁺ T cells with natural killer cell markers in comparison with natural killer cells and normal CD8⁺ T cells.

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We investigated the function of CD56⁺ CD8⁺ T cells (CD56⁺ T cells) and CD56⁻ CD57⁺ CD8⁺ T cells (CD57⁺ T cells; natural killer (NK)-type T cells) and compared them with those of normal CD56⁻ CD57⁻

CD8+ T cells (CD8+ T cells) and CD56+ NK cells from healthy volunteers. After the stimulation with immobilized anti-CD3 antibodies, both NK-type T cells produced much larger amounts of interferon-gamma (IFN-gamma) than CD8+ T cells. Both NK-type T cells also acquired a more potent cytotoxicity against NK-sensitive K562 cells than CD8+ T cells while only CD56+ T cells showed a potent cytotoxicity against NK-resistant Raji cells. After the stimulation with a combination of interleukin (IL)-2, IL-12 and IL-15, the IFN-gamma amounts produced were NK cells > or = CD56+ T cells > or = CD57+ T cells > CD8+ T cells. The cytotoxicities against K562 cells were NK cells > CD56+ T cells > or = CD57+ T cells > CD8+ T cells while cytotoxicities against Raji cells were CD56+ T cells > CD57+ T cells > or = CD8+ T cells > or = NK cells. However, the CD3-stimulated proliferation of both NK-type T cells was smaller than that of CD8+ T cells partly because NK-type T cells were susceptible to apoptosis. In addition to NK cells, NK-type T cells but not CD8+ T cells stimulated with cytokines, expressed cytoplasmic perforin and granzyme B. Furthermore, CD3-stimulated IFN-gamma production from peripheral blood mononuclear cells (PBMC) correlated with the proportions of CD57+ T cells in PBMC from donors. Our findings suggest that NK-type T cells play an important role in the T helper 1 responses and the immunological changes associated with ageing.

Granzyme B levels and granzyme B polymorphisms in peripheral blood of patients with endometriosis: a preliminary study.

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The chronic course of endometriosis suggests that the immune system may play a role in its aetiology. There may be resistance to cell lysis, as well as an immune defect underlying endometriosis. Granzyme B is a serine protease that is secreted by Natural Killer (NK) cells and cytotoxic T lymphocytes during a cellular immune response and can induce apoptosis. The aim of this study was to evaluate the relationship between both Granzyme B levels and Granzyme B gene polymorphisms in endometriosis patients. Women between the ages of 20 - 45 were included in the study. The patients were divided into two groups: those diagnosed with endometriosis and those who had not been diagnosed with endometriosis. In the blood samples, Granzyme B gene polymorphisms and serum levels of Granzyme B were studied. There was no difference between the groups in terms of median Granzyme B levels and the presence of AA, AG, and GG genotypes. There was a difference in median granzyme levels for the control group; the GG genotype was found at a lower frequency. The immune defect within endometriosis-related immune cells may not be exclusively due to Granzyme B. Other mediators that are secreted from immune cells may have additive effects.

IMPACT STATEMENT What is already known on this subject? NK cells are cytotoxic and inhibit the implantation of autologous endometrial cells that are spilled into the peritoneum by retrograde menstruation. Thus, a reduction in NK cell activity may facilitate the progression of endometriosis. The literature review reveals that there are studies suggesting that NK cell activity may be insufficient in endometriosis. Granzyme B is a serine protease that is secreted by NK cells and cytotoxic T lymphocytes during a cellular immune response. What do the results of this study add? Granzyme B is one of the cytotoxic granules in NK and cytotoxic T lymphocyte cells and its genetic polymorphisms were tested in endometriosis. We found that median Granzyme B levels were significantly different in patients with the GG genotype in the control group, compared to those with the AA and AG genotype. However, this difference was not detected between the control and endometriosis groups. What are the implications of these findings for clinical practice and/or further research? Our results contribute to uncovering the pathogenesis of endometriosis since there are no previous studies in the literature regarding this topic. Although we did not find a difference, our results will inform further studies made on this topic. Studies with different molecules and an increased number

of patients are needed. The immune defect of endometriosis may not be due exclusively to Granzyme B. Other mediators that are secreted from immune cells may have mutual effects and interactions.

In vivo rejection of tumor cells dependent on CD8 cells that kill independently of perforin and FasL.

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Perforin/granzyme B- and Fas/FasL-mediated killing pathways are the main effector mechanisms of CTL and NK cells in antitumor immune responses. In this study, we investigated the relative role of these two lytic mechanisms in protection of the host from tumor progression, as well as spontaneous metastasis, using the D122 Lewis lung carcinoma and its gene-modified cells. Utilizing perforin knockout mice (B6-PKO) and Fas and FasL mutant (B6-MRL and B6-Smn) mice, we found that perforin expression in the host plays a crucial function in the prevention of metastasis. However, local tumor rejection of an H-2K(b) and B7-1 transfectant, 39.5-B7 cells, was not dependent either on perforin or Fas/FasL expression in vivo. In addition, CTL lysis of 39.5-B7 cells was independent of perforin and Fas/FasL interactions in 18-hour in vitro assays. We also confirmed that CD8 T-cells were responsible for rejecting 39.5-B7 local tumors, yet cytokines, TNF-alpha and gammaIFN were not involved in tumor rejection in vivo. Furthermore, blocking assays using caspase inhibitors (zVAD-fmk, zLETD-fmk and zLEHD-fmk) showed that, whereas caspase activation was partially required to induce 39.5-B7 lysis mediated by the perforin-dependent pathway, 39.5-B7 lysis by CTLs through the perforin-independent mechanism required caspase activation. Thus, these results suggested that perforin, Fas/FasL, gammaIFN and TNF-alpha independent lytic mechanisms, mediated by CD8 T cells, have a crucial role in rejection of 39.5-B7 cells in vivo. Caspase activation is a pre requisite for apoptosis of targets by CTLs.

Genetic deletion of granzyme B does not confer resistance to the development of spontaneous diabetes in non-obese diabetic mice.

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Granzyme B (GzmB) and perforin are proteins, secreted mainly by natural killer cells and cytotoxic T lymphocytes that are largely responsible for the induction of apoptosis in target cells. Because type 1 diabetes results from the selective destruction of β cells and perforin deficiency effectively reduces diabetes in non-obese diabetic (NOD) mice, it can be deduced that β cell apoptosis involves the GzmB/perforin pathway. However, the relevance of GzmB remains totally unknown in non-obese diabetic (NOD) mice. In this study we have focused on GzmB and examined the consequence of GzmB deficiency in NOD mice. We found that NOD.GzmB(-/-) mice developed diabetes spontaneously with kinetics similar to those of wild-type NOD (wt-NOD) mice. Adoptive transfer study with regulatory T cell (Treg)-depleted splenocytes (SPCs) into NOD-SCID mice or in-vivo Treg depletion by anti-CD25 antibody at 4 weeks of age comparably induced the rapid progression of diabetes in the NOD.GzmB(-/-) mice and wt-NOD mice. Expression of GzmA and Fas was enhanced in the islets from pre-diabetic NOD.GzmB(-/-) mice. In contrast to spontaneous diabetes, GzmB deficiency suppressed the development of cyclophosphamide-promoted diabetes in male NOD mice. Cyclophosphamide treatment led to a significantly lower percentage of apoptotic CD4(+) , CD8(+) and CD4(+) CD25(+) T cells in SPCs from NOD.GzmB(-/-) mice than those from wt-NOD mice. In conclusion, GzmB, in contrast to perforin, is not

essentially involved in the effector mechanisms for β cell destruction in NOD mice.

B cell-derived circulating granzyme B is a feature of acute infectious mononucleosis.

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Granzyme B (GzmB) is a serine protease best known for inducing target cell apoptosis when released by cytotoxic T lymphocytes (CTLs) or natural killer cells with pore-forming perforin. As a result, GzmB detected in the serum of virus-infected individuals has typically been attributed to these sources. Here, we show that patients with recently diagnosed infectious mononucleosis caused by Epstein-Barr virus (EBV) have high circulating levels of GzmB that may be derived from infected B cells early in course of disease. We recently reported that human B cells from healthy donors secrete active GzmB when stimulated in vitro through B-cell receptor (BCR) ligation and interleukin (IL)-21. We found that infecting B cells with EBV greatly amplified GzmB secretion in response to the same stimuli, but the expression was terminated once the infection had become latent. Our results represent a rare instance of GzmB expression by non-CTL/natural killer cells in the context of infection with a human pathogen.

A possible role for perforin and granzyme B in resveratrol-enhanced radiosensitivity of prostate cancer.

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Perforin and granzyme B are expressed primarily by activated lymphocytes (cytotoxic T cells, natural killer cells, and natural killer T cells) and function together to induce apoptosis of target cells. Typically, these proteins are not expressed in tumor cells. In the present study, we established the constitutive expression of perforin and granzyme B by the PC-3 and DU145 prostate cancer (PCA) cell lines with reverse transcription polymerase chain reaction, immunohistochemistry, Western blot, or a combination of techniques. The combination of radiation and resveratrol (XRT/RSV) additively/synergistically decreased survival of PCA because, at least in part, of increased apoptosis. We further demonstrated that treatment with RSV up-regulated the expression of both perforin and granzyme B, whereas treatment with XRT up-regulated the expression of granzyme B, but not that of perforin. Combined XRT/RSV treatment of PCA cells further increased the expression of both perforin and granzyme B compared with RSV or XRT alone. Thus, increased radiosensitivity of prostate cancer cells induced by RSV correlated with up-regulation of perforin and granzyme B, demonstrating a possible mechanism for tumor apoptosis. These findings might be helpful in devising new strategies for treating PCA.

The transcriptional control of the perforin locus.

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Natural killer (NK) cells and cytotoxic T lymphocytes (CTLs) use cytotoxic granules containing perforin and granzymes to lyse infected or malignant host cells, thereby providing immunity to intracellular microbes and tumors. Perforin is essential for cytotoxic granule-mediated killing. Perforin expression is regulated transcriptionally and correlates tightly with the development of cells that can exhibit cytotoxic

activity. Although a number of genes transcribed by T cells and NK cells have been studied, the cell-specificity of perforin gene expression makes it an ideal model system in which to clarify the transcriptional mechanisms that guide the development and activation of cytotoxic lymphocytes. In this review, we discuss what is known about perforin expression and its regulation, then elaborate on recent studies that utilized chromosome transfer and bacterial artificial chromosome transgenics to define a comprehensive set of cis-regulatory regions that control transcription of the human PRF1 gene in a near-physiologic context. In addition, we compare the human and murine Prf1 loci and discuss how transcription factors known to be important for driving CTL differentiation might also directly regulate the cis-acting domains that control Prf1. Our review emphasizes how studies of PRF1/Prf1 gene transcription can illuminate not only the mechanisms of cytotoxic lymphocyte differentiation but also some basic principles of transcriptional regulation.

Channel formation and $[Ca^{2+}]_i$ accumulation induced by perforin N-terminus peptides: comparison with purified perforin and whole lytic granules.

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Cytotoxic T lymphocytes (CTL) and natural killer (NK) cells express the pore forming protein perforin, which contributes to lymphocytotoxicity. The hallmark of perforin action is opening high-conductance transmembrane channels that enable massive influx of Ca^{2+} ions (deleterious to many cell types), as well as granzymes, which may trigger the apoptotic pathway. To explore the functional domains in the perforin molecule, we investigated in PN71 lymphocytes, the ability of perforin N-terminus synthetic peptides (compared to purified perforin and perforin-containing lytic granules), to cause intracellular Ca^{2+} ($[Ca^{2+}]_i$) accumulation and open transmembrane channels. To this end, we used the whole cell recording technique and Indo 1 fluorescence to measure membrane currents and $[Ca^{2+}]_i$, respectively. We have demonstrated that the N-terminus peptide Hu-34 (amino acids 1-34) closely resembled perforin action, reflected by $[Ca^{2+}]_i$ accumulation and channel activity, while shorter peptides (e.g., Hu-16) generated mostly short-lived channels but no $[Ca^{2+}]_i$ elevation. Hence, the first 34 amino acids of the perforin N-terminus sequence are sufficient for the perforin action.